

AN ALTERNATIVE TO ‘THE LOGARITHMIC-WITH-ZEROS DISTRIBUTION’ AND ITS GENERALIZATION: PROPERTIES AND ESTIMATION

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Summary: Through this article we propose a class of distributions namely “the alpha-logarithmic series distribution (ALSD)” as an alternative to “the logarithmic-with-zeros distribution” of Khatri (1961) and develop a generalization of it and call it “the generalized alpha-logarithmic series distribution (GALSD)”. Several properties of these classes of distributions are studied here. The maximum likelihood estimation of the parameters of the GALSD is discussed and fitted to certain real life data sets for illustrating its usefulness compared to the existing models. Further, the generalized likelihood ratio test is considered for testing the significance of the additional parameter of the GALSD model and a simulation study is carried out for assessing the performance of the estimators of each parameter of the model.

1. Introduction

Fischer, Corbert and Williams (1943) considered the logarithmic series distribution (LSD) as a limiting form of the zero-truncated negative binomial distribution. A random variable Z is said to follow the logarithmic series distribution (LSD) if its probability mass function (pmf) is the following:

$$f_1(z) = \frac{A\theta^z}{z} \quad (1)$$

for $z = 1, 2, 3, \dots$ in which $A = [-\ln(1 - \theta)]^{-1}$ and $0 < \theta < 1$. The probability generating function (pgf) of the LSD with pmf (1) is

$$P(t) = -A \ln(1 - \theta t), \quad (2)$$

which can also be written as

$$P(t) = \frac{{}_2F_1(1, 1; 2; \theta t)}{{}_2F_1(1, 1; 2; \theta)}$$

where

$${}_2F_1(a, b; c; \theta) = \sum_{r=0}^{\infty} \frac{(a)_r (b)_r}{(c)_r} \frac{\theta^r}{r!}$$

is the Gauss hypergeometric function, in which $(d)_r = d(d+1)(d+2) \cdots (d+r-1)$ for $r \geq 1$ and $(d)_0 = 1$ for any $d \in R = (-\infty, \infty)$. For details regarding Gauss hypergeometric function see Mathai and Haubold (2008) or Slater (1966)

The LSD has found applications in several areas of research such as biology, ecology, economics and marine sciences. Consequently, generalized forms of the LSD have received much attention in the literature. For example see Ong (2000), Khang and Ong (2007) and Kumar and Riyaz (2013a, 2013b, 2013c). An important drawback of the LSD in the practical point of view is that it excludes zero observations from its support. To overcome this practical difficulty, Khatri (1961) considered a distribution namely “the logarithmic-with-zeros distribution (LWZD)” through the following probability mass function (pmf):

$$f_2(z) = \begin{cases} \omega, & z = 0 \\ \frac{A(1-\omega)\theta^z}{z}, & z = 1, 2, 3, \dots \end{cases} \quad (3)$$

Some aspects of the LWZD are also detailed in Johnson, Kemp and Kotz (2005, p. 355).

In the present paper, our motive is to propose an alternative form of the LWZD with name “the alpha-logarithmic series distribution (ALSD)” and to study some of its important aspects. In Section 2 we present the definition of the ALSD and derive its important properties. In Section 3, we consider a generalized version of the ALSD and call it “the generalized alpha-logarithmic series distribution (GALSD)”, and discuss its important properties. The maximum likelihood estimation of the parameters of the GALSD is attempted in Section 4 and its usefulness is illustrated with the help of certain real life data sets. In Section 5 we consider the generalized likelihood ratio test for testing the significance of the additional parameter of the GALSD and in Section 6, we carried out a simulation study for comparing the performance of the estimators obtained by the method of maximum likelihood.

We need the following series representations in the sequel. For any real valued function $Q(r, s)$,

$$\sum_{s=0}^{\infty} \sum_{r=0}^{\infty} Q(r, s) = \sum_{s=0}^{\infty} \sum_{r=0}^s Q(r, s-r) \quad (4)$$

$$= \sum_{s=0}^{\infty} \sum_{r=0}^{\left[\frac{s}{2}\right]} Q(r, s-2r), \quad (5)$$

in which $[a]$ denotes the integer part of a , for any $a > 0$ and for any $b > 0$ and $|u| < 1$,

$$(1-u)^{-b} = \sum_{r=0}^{\infty} (b)_r \frac{u^r}{r!}. \quad (6)$$

2. Alpha-logarithmic series distribution

In this section first we present the definition of the ALSD and discuss some of its properties.

Definition 1 A non-negative integer valued random variable is said to follow the alpha-logarithmic series distribution (ALSD) if its pmf $g_x = P(X = x)$ is the following:

$$g_x = \begin{cases} -B \ln(1 - \alpha), & x = 0 \\ B(1 - \alpha)^{-x} \frac{\theta^x}{x}, & x = 1, 2, 3, \dots \end{cases} \quad (7)$$

in which $B^{-1} = -\ln(1 - \theta - \alpha)$, $\theta > 0$, $\alpha \geq 0$ such that $\theta + \alpha < 1$.

Clearly, when $\alpha = 0$ the pmf (7) reduces to the pmf of the standard LSD with pmf as given in (1). Now we derive the pgf of the ALSD through the following result.

Proposition 1 The pgf $G(\cdot)$ of the ALSD with pmf (7) is the following:

$$G(t) = -B \ln(1 - \theta t - \alpha). \quad (8)$$

Proof. By the definition of pgf we have,

$$\begin{aligned} G(t) &= \sum_{x=0}^{\infty} g_x t^x \\ &= B \sum_{x=1}^{\infty} (1 - \alpha)^{-x} \frac{\theta^x}{x} t^x - B \ln(1 - \alpha) \\ &= B \sum_{x=1}^{\infty} \sum_{r=0}^{\infty} (x)_r \frac{\alpha^r}{r!} \frac{\theta^x}{x} t^x - B \ln(1 - \alpha) \end{aligned} \quad (9)$$

by (6). Now writing $(x)_r = \frac{(x+r-1)!}{(x-1)!}$ in (9) we get

$$\begin{aligned} G(t) &= B \sum_{x=1}^{\infty} \sum_{r=0}^{\infty} \frac{(x+r-1)!}{x! r!} \alpha^r (\theta t)^x - B \ln(1 - \alpha) \\ &= B \sum_{x=0}^{\infty} \sum_{r=0}^{\infty} \frac{1}{(x+r+1)} \binom{x+r+1}{r} \alpha^r (\theta t)^{x+1} - B \ln(1 - \alpha). \end{aligned} \quad (10)$$

By applying (4) in (10) and expanding $[-\ln(1 - \alpha)]$ we obtain the following.

$$\begin{aligned} G(t) &= B \sum_{x=0}^{\infty} \sum_{r=0}^x \frac{1}{(x+1)} \binom{x+1}{r} \alpha^r (\theta t)^{x-r+1} - B \sum_{x=0}^{\infty} \frac{\alpha^{x+1}}{x+1} \\ &= B \sum_{x=0}^{\infty} \sum_{r=0}^{x+1} \frac{1}{(x+1)} \binom{x+1}{r} \alpha^r (\theta t)^{x-r+1} \\ &= B \sum_{x=0}^{\infty} \frac{(\theta t + \alpha)^{x+1}}{(x+1)} \end{aligned} \quad (11)$$

in the light of the binomial theorem. Thus we have

$$G(t) = B \sum_{x=1}^{\infty} \frac{(\theta t + \alpha)^x}{x},$$

which gives (8). ■

Proposition 2 For $r \geq 1$, an expression for the r^{th} factorial moment of the ALSD is

$$\mu_{[r]} = B(r-1)! \theta^r (1 - \theta - \alpha)^{-r}. \quad (12)$$

Proof. By definition, the factorial moment generating function $F(\cdot)$ of the ALSD with pgf (8) is

$$\begin{aligned} F(t) &= \sum_{r=0}^{\infty} \mu_{[r]} \frac{t^r}{r!} \\ &= G(1+t) \\ &= -B \ln[1 - \theta(1+t) - \alpha] \\ &= -B \ln\left\{(1-\alpha)\left[1 - \frac{\theta}{1-\alpha}(1+t)\right]\right\} \\ &= -B\left\{\ln(1-\alpha) + \ln\left[1 - \frac{\theta}{1-\alpha} - \frac{\theta}{1-\alpha}t\right]\right\} \\ &= -B\left\{\ln(1-\alpha) + \ln\left(1 - \frac{\theta}{1-\alpha}\right) + \ln\left(1 - \frac{\theta}{1-\theta-\alpha}t\right)\right\}. \end{aligned} \quad (13)$$

On expanding the third logarithmic term in (13) and equating the coefficients of $\frac{t^r}{r!}$ we obtain (12). ■

Remark 1 The pgf given in (8) has the following representation in terms of the Gauss hypergeometric function.

$$G(t) = \frac{\theta t + \alpha}{\theta + \alpha} \frac{{}_2F_1(1, 1; 2; \theta t + \alpha)}{{}_2F_1(1, 1; 2; \theta + \alpha)}. \quad (14)$$

Remark 2 If we replace t by e^{it} in (8) we get the characteristic function $\phi(t)$ of the ALSD as given below, in which $t \in R$ and $i = \sqrt{-1}$,

$$\phi(t) = -B \ln(1 - \theta e^{it} - \alpha). \quad (15)$$

Proposition 3 The mean and variance of the ALSD with pgf (8) are

$$\text{mean} = \frac{B\theta}{1 - \theta - \alpha}$$

and

$$\text{variance} = \frac{B\theta(1 - B\theta - \alpha)}{(1 - \theta - \alpha)^2}.$$

Proof. The proof follows from the fact that

$$\text{mean} = G^{(1)}(1)$$

and

$$\text{variance} = G^{(2)}(1) + G^{(1)}(1) - (G^{(1)}(1))^2$$

where

$$G^{(r)}(t) = \frac{d^r G(t)}{dt^r} / t = 1.$$

■

Proposition 4 For $r \geq 1$, the r^{th} raw moment μ'_r of the ALSD with characteristic function (15) is

$$\mu'_r = B\delta\phi(\delta, 1-r, 1) \quad (16)$$

in which $\delta = (1-\alpha)^{-1}\theta$ and

$$\phi(\delta, 1-r, 1) = \sum_{j=0}^{\infty} (j+1)^{r-1} \delta^j \quad (17)$$

is the Lerch function for any real δ and $r \geq 1$ (Johnson et al., 2005, p. 20)

Proof. From (15) we have

$$\phi(t) = \sum_{r=1}^{\infty} \mu'_r \frac{(it)^r}{r!} \quad (18)$$

$$= -B \ln(1 - \theta e^{it} - \alpha). \quad (19)$$

On expanding the logarithmic function in (19) we get

$$\begin{aligned} \phi(t) &= B \sum_{j=0}^{\infty} \frac{(\theta e^{it} + \alpha)^{j+1}}{j+1} \\ &= B \sum_{j=0}^{\infty} \sum_{k=0}^{j+1} \frac{j!}{k!(j-k+1)!} (\theta e^{it})^{j-k+1} \alpha^k, \end{aligned} \quad (20)$$

by the binomial theorem. Now by separating the last term of the inside summation of (20), we have

$$\phi(t) = B \sum_{j=0}^{\infty} \sum_{k=0}^j \frac{j!}{k!(j-k+1)!} (\theta e^{it})^{j-k+1} \alpha^k + B \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1}. \quad (21)$$

Apply (4) in the first term of (21) to get the following:

$$\begin{aligned} \phi(t) &= B \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{(j+k)!}{k!(j+1)!} (\theta e^{it})^{j+1} \alpha^k + B \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1}, \\ &= B \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{(j+k)!}{j!} \frac{\alpha^k}{k!} \frac{\theta^{j+1}}{j+1} \sum_{r=0}^{\infty} (j+1)^r \frac{(it)^r}{r!} + B \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1} \end{aligned} \quad (22)$$

by expanding the exponential term and writing the identity $(j+k)! = j!(1+j)_k$. Now, on applying (6) in (22) we get

$$\phi(t) = B \sum_{r=0}^{\infty} \sum_{j=0}^{\infty} \frac{\left(\frac{\theta}{1-\alpha}\right)^{j+1}}{j+1} (j+1)^r \frac{(it)^r}{r!} + B \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1}. \quad (23)$$

On equating the coefficients of $\frac{(it)^r}{r!}$ on right hand side expressions of (18) and (23), we get (16), in the light of (17). ■

3. Generalized alpha-logarithmic series distribution

In this section we define a generalized form of the ALS D and discuss some of its important properties.

Definition 2 A non-negative integer valued random variable Y is said to follow “the generalized alpha-logarithmic series distribution (GALSD)” if its pgf is of the following form, in which, $C^{-1} = -\ln(1 - \theta_1 - \theta_2 - \alpha)$, $\theta_1 > 0$, $\theta_2 \geq 0$ and $\alpha \geq 0$ such that $\theta_1 + \theta_2 + \alpha < 1$,

$$H(t) = -C \ln(1 - \theta_1 t - \theta_2 t^2 - \alpha). \quad (24)$$

Clearly, when $\theta_2 = 0$ the pgf (24) reduces to the pgf of the ALS D as given in (8) with $\theta = \theta_1$ and when $\theta_2 = 0$ and $\alpha = 0$ the pgf (24) reduces to the pgf of the standard LSD as given in (2) with $\theta = \theta_1$. When $\alpha = 0$ the pgf (24) reduces to

$$H(t) = \frac{\ln(1 - \theta_1 t - \theta_2 t^2)}{\ln(1 - \theta_1 - \theta_2)}. \quad (25)$$

Note that a distribution with the pgf (25) can be considered a special case of the modified logarithmic series distribution of Wimmer and Altmann (1995), which they obtained as the limiting case of the zero-truncated form of the generalized Gegenbauer distribution studied by Medhi and Borah (1984). The pgf of the GALSD given in (24) can also be written as

$$H(t) = \frac{(\theta_1 t + \theta_2 t^2 + \alpha)}{(\theta_1 + \theta_2 + \alpha)} \frac{{}_2F_1(1, 1; 2; \theta_1 t + \theta_2 t^2 + \alpha)}{{}_2F_1(1, 1; 2; \theta_1 + \theta_2 + \alpha)}. \quad (26)$$

If we replace t by e^{it} in (24) we get the characteristic function $\phi(t)$ of the GALSD as given below,

$$\phi(t) = -C \ln(1 - \theta_1 e^{it} - \theta_2 e^{2it} - \alpha). \quad (27)$$

We obtain the pmf of the GALSD through the following result.

Proposition 5 The pmf $h_x = P(Y = x)$ of the GALSD with pgf (24) is the following:

$$h_x = \begin{cases} -C \ln(1 - \alpha), & x = 0 \\ C \sum_{r=0}^{\lfloor \frac{x}{2} \rfloor} \frac{(x-r-1)!(1-\alpha)^{-(x-r)} \theta_1^{x-2r} \theta_2^r}{(x-2r)! r!}, & x = 1, 2, 3, \dots \end{cases} \quad (28)$$

Proof. From (24) we have the following:

$$H(t) = \sum_{x=0}^{\infty} h_x t^x \quad (29)$$

$$= -C \ln(1 - \theta_1 t - \theta_2 t^2 - \alpha). \quad (30)$$

On expanding the logarithmic function in (30), we get

$$\begin{aligned} H(t) &= C \sum_{x=0}^{\infty} \frac{(\theta_1 t + \theta_2 t^2 + \alpha)^{x+1}}{x+1} \\ &= C \sum_{x=0}^{\infty} \sum_{p=0}^{x+1} \frac{x!}{p!(x-p+1)!} (\theta_1 t + \theta_2 t^2)^{x-p+1} \alpha^p, \end{aligned} \quad (31)$$

by the binomial theorem. Now (31) can be written as

$$H(t) = C \sum_{x=0}^{\infty} \sum_{p=0}^x \frac{x!}{p!(x-p+1)!} (\theta_1 t + \theta_2 t^2)^{x-p+1} \alpha^p + C \sum_{x=0}^{\infty} \frac{\alpha^{x+1}}{(x+1)!}. \quad (32)$$

By applying the binomial theorem in the first term and since the second term is the logarithmic expansion of $[-\ln(1-\alpha)]$, we obtain the following from (32):

$$H(t) = C \sum_{x=0}^{\infty} \sum_{p=0}^x \sum_{r=0}^{x-p+1} \frac{x!}{p!(x-p+1)!} \binom{x-p+1}{r} \theta_1^{x-p-r+1} \theta_2^r \alpha^p t^{x-p+r+1} - C \ln(1-\alpha). \quad (33)$$

Applying (4) in (33) to get

$$H(t) = C \sum_{x=0}^{\infty} \sum_{p=0}^{\infty} \sum_{r=0}^{x+1} \frac{(x+p)!}{p!(x+1)!} \binom{x+1}{r} \theta_1^{x-r+1} \theta_2^r \alpha^p t^{x+r+1} - C \ln(1-\alpha). \quad (34)$$

On splitting the first term of (34), we have

$$\begin{aligned} H(t) = & C \sum_{x=0}^{\infty} \sum_{p=0}^{\infty} \sum_{r=0}^x \frac{(x+p)!}{p!(x+1)!} \binom{x+1}{r} \theta_1^{x-r+1} \theta_2^r \alpha^p t^{x+r+1} \\ & + C \sum_{x=0}^{\infty} \sum_{p=0}^{\infty} \frac{(x+p)!}{p!(x+1)!} \theta_2^{x+1} \alpha^p t^{2(x+1)} - C \ln(1-\alpha). \end{aligned} \quad (35)$$

By applying (4) in (35) we get

$$\begin{aligned} H(t) = & C \sum_{x=0}^{\infty} \sum_{p=0}^{\infty} \sum_{r=0}^{\infty} \frac{(x+p+r)!}{p!(x+r+1)!} \binom{x+r+1}{r} \theta_1^{x+1} \theta_2^r \alpha^p t^{x+2r+1} \\ & + C \sum_{x=0}^{\infty} \sum_{p=0}^{\infty} \frac{(x+p)!}{p!(x+1)!} (\theta_2 t^2)^{x+1} \alpha^p - C \ln(1-\alpha), \end{aligned} \quad (36)$$

which implies the following in the light of (5).

$$\begin{aligned} H(t) = & C \sum_{x=0}^{\infty} \sum_{p=0}^{\infty} \sum_{r=0}^{\lfloor \frac{x}{2} \rfloor} \frac{(x-r+p)!}{p!(x-r+1)!} \binom{x-r+1}{r} \theta_1^{x-2r+1} \theta_2^r \alpha^p t^{x+1} \\ & + C \sum_{x=0}^{\infty} \sum_{p=0}^{\infty} \frac{(x+p)!}{p!(x+1)!} (\theta_2 t^2)^{x+1} \alpha^p - C \ln(1-\alpha). \end{aligned} \quad (37)$$

$$\begin{aligned} = & C \sum_{x=0}^{\infty} \sum_{r=0}^{\lfloor \frac{x}{2} \rfloor} (x-r)! (1-\alpha)^{-(1+x-r)} \frac{\theta_1^{x-2r+1}}{(x-2r+1)!} \frac{\theta_2^r}{r!} t^{x+1} \\ & + C \sum_{x=0}^{\infty} (1-\alpha)^{-(1+x)} \frac{(\theta_2 t^2)^{x+1}}{x+1} - C \ln(1-\alpha), \end{aligned} \quad (38)$$

by (6). Now, on equating the coefficients of t^x on the right hand side expressions of (29) and (38) we get (28). ■

Proposition 6 The mean and variance of the GALSD are

$$\text{mean} = \frac{C(\theta_1 + 2\theta_2)}{1 - \theta_1 - \theta_2 - \alpha}$$

and

$$\text{variance} = \frac{C}{1 - \theta_1 - \theta_2 - \alpha} [(\theta_1 + 4\theta_2) + (1 - C) \frac{(\theta_1 + 2\theta_2)^2}{1 - \theta_1 - \theta_2 - \alpha}].$$

Proof is similar to that of Proposition 2.

For the computation of higher order moments of the GALSD, we derive the following result.

Proposition 7 For $r \geq 1$, the r^{th} raw moment μ'_r of the GALSD with characteristic function (27) is the following, in which $\delta_j = (1 - \alpha)^{-1} \theta_j$, for $j = 1, 2$

$$\mu'_r = CS(\delta_1, \delta_2; r) + 2^r C \delta_2 \phi(\delta_2, 1 - r, 1) \quad (39)$$

where

$$S(\delta_1, \delta_2; r) = \sum_{j=0}^{\infty} \sum_{m=0}^{\infty} (j+m)! (j+2m+1)^r \frac{\delta_1^{j+1}}{(j+1)!} \frac{\delta_2^m}{m!}$$

which can be computed for particular values of δ_1, δ_2 and r using some mathematical softwares such as MATHEMATICA, MATHCAD etc., since the series converges for the parametric restriction imposed in Definition 2 and

$$\phi(\delta_2, 1 - r, 1) = \sum_{j=0}^{\infty} (j+1)^{r-1} \delta_2^j$$

is the Lerch function, as defined in (17).

Proof. From (27) we have the following, in which $t \in R$ and $i^2 = -1$,

$$\phi(t) = \sum_{r=1}^{\infty} \mu'_r \frac{(it)^r}{r!} \quad (40)$$

$$= -C \ln(1 - \theta_1 e^{it} - \theta_2 e^{2it} - \alpha). \quad (41)$$

On expanding the logarithmic function in (41), we get

$$\begin{aligned} \phi(t) &= C \sum_{j=0}^{\infty} \frac{(\theta_1 e^{it} + \theta_2 e^{2it} + \alpha)^{j+1}}{j+1} \\ &= C \sum_{j=0}^{\infty} \sum_{k=0}^{j+1} \frac{j!}{k!(j-k+1)!} (\theta_1 e^{it} + \theta_2 e^{2it})^{j-k+1} \alpha^k, \end{aligned} \quad (42)$$

by the binomial theorem. Now (42) can be split as

$$\phi(t) = C \sum_{j=0}^{\infty} \sum_{k=0}^j \frac{j!}{k!(j-k+1)!} (\theta_1 e^{it} + \theta_2 e^{2it})^{j-k+1} \alpha^k + C \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1}. \quad (43)$$

By applying the binomial theorem in the first term of (43), we get the following:

$$\phi(t) = C \sum_{j=0}^{\infty} \sum_{k=0}^j \sum_{m=0}^{j-k+1} \frac{j!}{k!(j-k+1)!} \binom{j-k+1}{m} \theta_1^{j-k-m+1} \theta_2^m \alpha^k e^{it(j-k+m+1)} + C \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1} \quad (44)$$

$$= C \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \sum_{m=0}^{j+1} \frac{(j+k)!}{k!(j+1)!} \binom{j+1}{m} \theta_1^{j-m+1} \theta_2^m \alpha^k e^{it(j+m+1)} + C \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1} \quad (45)$$

in the light of (4). Now, splitting the summation over m in (45) to obtain

$$\begin{aligned} \phi(t) = & C \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \sum_{m=0}^j \frac{(j+k)!}{k!(j+1)!} \binom{j+1}{m} \theta_1^{j-m+1} \theta_2^m \alpha^k e^{it(j+m+1)} \\ & + C \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{(j+k)!}{k!(j+1)!} \theta_2^{j+1} \alpha^k e^{2it(j+1)} + C \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1}. \end{aligned} \quad (46)$$

Apply (4) in the first term of (46) to get the following:

$$\begin{aligned} \phi(t) = & C \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} \frac{(j+m+k)!}{k!(j+m+1)!} \binom{j+m+1}{m} \theta_1^{j+1} \theta_2^m \alpha^k e^{it(j+2m+1)} \\ & + C \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{(j+k)!}{k!(j+1)!} \theta_2^{j+1} \alpha^k e^{2it(j+1)} + C \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1}. \end{aligned} \quad (47)$$

On expanding the exponential terms in (47), we get

$$\begin{aligned} \phi(t) = & C \sum_{j=0}^{\infty} \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} (j+m+k)! \frac{\alpha^k}{k!} \frac{\theta_1^{j+1}}{(j+1)!} \frac{\theta_2^m}{m!} \sum_{r=0}^{\infty} (j+2m+1)^r \frac{(it)^r}{r!} \\ & + C \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{(j+k)!}{j!} \frac{\alpha^k}{k!} \frac{\theta_2^{j+1}}{(j+1)!} \sum_{r=0}^{\infty} (j+1)^r \frac{(2it)^r}{r!} + C \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1}. \end{aligned} \quad (48)$$

On writing $(j+m+k)! = (j+m)!(1+j+m)_k$ in the first term of (48) and $(j+k)! = j!(1+j)_k$ in the second term of (48) we get the following in the light of (6):

$$\begin{aligned} \phi(t) = & C \sum_{r=0}^{\infty} \sum_{j=0}^{\infty} \sum_{m=0}^{\infty} (j+m)!(1-\alpha)^{-(1+j+m)} (j+2m+1)^r \frac{\theta_1^{j+1}}{(j+1)!} \frac{\theta_2^m}{m!} \frac{(it)^r}{r!} \\ & + C \sum_{r=0}^{\infty} \sum_{j=0}^{\infty} (1-\alpha)^{-(1+j)} (j+1)^r \frac{\theta_2^{j+1}}{(j+1)!} \frac{(2it)^r}{r!} + C \sum_{j=0}^{\infty} \frac{\alpha^{j+1}}{j+1}. \end{aligned} \quad (49)$$

On equating the coefficients of $\frac{(it)^r}{r!}$ in the right hand side expressions of (40) and (49) we get (39). ■

4. Estimation

In this section we discuss the estimation of the parameters of the GALSD by the method of maximum likelihood and illustrate the procedure by using certain real life data sets.

Let $a(x)$ be the observed frequency of x events and let y be the highest value of x observed. Then the likelihood function of the sample is

$$\begin{aligned} L &= \prod_{x=0}^y h_x^{a(x)} \\ &= h_0^{a(0)} \prod_{x=0}^y h_x^{a(x)} \end{aligned} \quad (50)$$

where h_x is the pmf of the GALSD as given in (28). Taking logarithm on both sides of (50), we have

$$\begin{aligned} \log L &= a(0) \log h_0 + \sum_{x=1}^y a(x) \log(h_x) \\ &= a(0) \log h_0 + \sum_{x=1}^y a(x) [\log C + \log \psi(x; \theta_1; \theta_2; \alpha)], \end{aligned} \quad (51)$$

where C is as defined in (24) and

$$\psi(x; \theta_1; \theta_2; \alpha) = \sum_{r=0}^{\left[\frac{x}{2}\right]} (x-r-1)! (1-\alpha)^{-(x-r)} \frac{\theta_1^{x-2r}}{(x-2r)!} \frac{\theta_2^r}{r!}.$$

Let $\hat{\theta}_1, \hat{\theta}_2$ and $\hat{\alpha}$ denote the maximum likelihood estimators of the parameters θ_1, θ_2 and α respectively of the GALSD. On differentiating (51) partially with respect to the parameters θ_1, θ_2 and α respectively and equating to zero, we get the following likelihood equations, in which

$$D = \xi(\theta_1, \theta_2, \alpha) = -C(1 - \theta_1 - \theta_2 - \alpha)^{-1}$$

$$a(0)D + \sum_{x=1}^y a(x) \left[D + \frac{\sum_{r=0}^{\left[\frac{x}{2}\right]} (x-r-1)! (1-\alpha)^{-(x-r)} \frac{\theta_1^{x-2r-1}}{(x-2r-1)!} \frac{\theta_2^r}{r!}}{\psi(x; \theta_1; \theta_2; \alpha)} \right] = 0 \quad (52)$$

$$a(0)D + \sum_{x=1}^y a(x) \left[D + \frac{\sum_{r=1}^{\left[\frac{x}{2}\right]} (x-r-1)! (1-\alpha)^{-(x-r)} \frac{\theta_1^{x-2r}}{(x-2r)!} \frac{\theta_2^{r-1}}{(r-1)!}}{\psi(x; \theta_1; \theta_2; \alpha)} \right] = 0 \quad (53)$$

and

$$a(0) \left[\frac{1}{-\ln(1-\alpha)(1-\alpha)} + D \right] + \sum_{x=1}^y a(x) \left[D + \frac{\sum_{r=0}^{\left[\frac{x}{2}\right]} (x-r)! (1-\alpha)^{-(x-r-1)} \frac{\theta_1^{x-2r}}{(x-2r)!} \frac{\theta_2^r}{r!}}{\psi(x; \theta_1; \theta_2; \alpha)} \right] = 0. \quad (54)$$

When likelihood equations do not have a solution the maximum of the likelihood function attained at the border of the domain of parameters. So we obtained the second order partial derivatives of $\log h_x$ with respect to parameters θ_1, θ_2 and α and by using MATHCAD software we observed that these equations give negative values for all $\theta_1 > 0, \theta_2 \geq 0$ and $\alpha \geq 0$ such that $\theta_1 + \theta_2 + \alpha < 1$. Thus the density of the GALSD is log-concave and hence the maximum likelihood estimators of the parameters θ_1, θ_2 and α are unique (cf. Puig, 2003). Now, on solving these likelihood equations (52), (53) and (54) by using some mathematical software such as MATHLAB, MATHCAD, MATHEMATICA etc. one can obtain the maximum likelihood estimators of the parameters θ_1, θ_2 and α of the GALSD.

For numerical illustration, we have considered two real life data sets, of which the first data set is on the numbers of diseased plants in groups of 9 plants, taken from Cochran (1954) and the second data set is on observed frequencies of *Pyrausta Nubilais*, taken from McGuire, Brindley and Bancroft (1952). We have fitted the zero-inflated Poisson distribution (ZIPD), zero-inflated negative binomial distribution (ZINBD), the LWZD, the ALSZ and the GALSD to both the data sets and the results thus obtained along with the corresponding values of the expected frequencies, Chi-square statistic, degrees of freedom (d.f.), P, Akaike information criterion (AIC), Bayesian information criterion (BIC) and the second order Akaike information criterion (AICc) in respect of each of the models are presented in Tables 1 and 2. Based on the computed values of Chi-square statistic, P, AIC, BIC and AICc listed in the tables, it can be observed that the GALSD gives a better fit to both the data sets compared to the existing models.

Table 1: Observed frequencies and computed values of expected frequencies of the ALSZ, the ZIPD, the ZINBD, the LWZD and the GALSD by the method of maximum likelihood for the first data set.

x	Observed	ALSD	ZIPD	LWZD	ZINBD	GALSD
0	36	42.854	68.160	72.160	65.605	38.398
1	48	43.262	35.247	54.386	32.469	50.37
2	38	20.009	29.696	17.676	21.201	37.313
3	23	12.339	16.679	7.659	13.867	19.069
4	10	8.560	7.026	3.734	9.077	9.404
5	3	6.334	2.368	1.942	5.945	3.476
6	1	4.883	0.665	1.052	3.895	1.101
7	1	3.871	0.155	0.586	2.552	0.012
8	0	21.888	0.004	0.805	5.389	0.857
Total	160	160	160	160	160	160
Estimates of the parameters		$\hat{\theta}_1 = 0.37$ $\hat{\alpha} = 0.60$	$\hat{\theta} = 1.685$ $\hat{\omega} = 0.295$	$\hat{\theta} = 0.650$ $\hat{\omega} = 0.451$	$\hat{\rho} = 0.101$ $\hat{k} = 0.99$ $\hat{\mu} = 1.89$	$\hat{\theta}_1 = 0.31$ $\hat{\theta}_2 = 0.16$ $\hat{\alpha} = 0.29$
Chi-square values		55.773	25.098	50.057	51.659	1.213
d.f.		5	2	2	4	2
P.values		< 0.0001	< 0.0001	< 0.0001	< 0.0001	0.545
AIC		263.008	243.234	248.434	256.078	226.426
BIC		263.416	243.642	248.842	256.690	227.038
AICc		263.441	243.667	248.867	256.232	226.580

Table 2: Observed frequencies and computed values of expected frequencies of the ALS D, the ZIP D, the ZINBD, the LWZD and the GALSD by the method of maximum likelihood for the second data set.

k	Observed	ALS D	ZIP D	LWZD	ZINBD	GALSD
0	89	94.779	107.892	113.724	97.376	89.433
1	96	90.333	90.112	93.104	58.461	95.818
2	57	40.534	69.927	39.802	48.532	60.213
3	44	34.251	36.175	22.687	38.289	41.143
4	16	16.323	14.437	14.548	29.419	16.696
5	11	11.719	4.357	9.951	22.240	9.711
6	7	8.764	1.127	7.090	16.630	6.132
7	3	6.742	0.250	5.196	12.337	3.920
8	1	20.555	0.124	17.898	0.716	0.934
Total	324	324	324	324	324	324
Estimates of the parameters		$\hat{\theta} = 0.35$ $\hat{\alpha} = 0.61$	$\hat{\theta} = 1.552$ $\hat{\omega} = 0.154$	$\hat{\theta} = 0.855$ $\hat{\omega} = 0.351$	$\hat{p} = 0.01$ $\hat{k} = 1.35$ $\hat{\mu} = 3.25$	$\hat{\theta}_1 = 0.30$ $\hat{\theta}_2 = 0.14$ $\hat{\alpha} = 0.35$
Chi-square values		33.209	52.530	50.057	48.251	0.534
d.f.		6	3	6	4	3
P.values		< 0.0001	< 0.0001	< 0.0001	< 0.0001	0.911
AIC		516.342	510.720	591.361	518.023	496.023
BIC		517.363	511.741	592.380	519.551	497.552
AICc		516.379	510.762	591.393	518.071	496.071

5. Testing of the hypothesis

In this section we discuss the testing of the hypothesis $H_0 : \theta_2 = 0$ against the alternative hypothesis $H_1 : \theta_2 \neq 0$ by using generalized likelihood ratio test.

In case of generalized likelihood ratio test, the test statistic is

$$-2 \log \lambda = 2[\log L(\hat{\underline{\Omega}}; x) - \log L(\hat{\underline{\Omega}}^*; x)], \quad (55)$$

where $\hat{\underline{\Omega}}$ is the maximum likelihood estimator of $\underline{\Omega} = (\theta_1, \theta_2, \alpha)$ with no restrictions, $\hat{\underline{\Omega}}^*$ and is the maximum likelihood estimator of $\underline{\Omega}$ when $\theta_2 = 0$. The test statistic $-2 \log \lambda$ given in (55) is asymptotically distributed as χ^2 with one degree of freedom. For details see Rao (1973). We have computed the values of $\log L(\hat{\underline{\Omega}}; x)$, $\log L(\hat{\underline{\Omega}}^*; x)$ and the test statistic for the GALSD for these three data sets and presented in Table 3. Since the critical value for the test at 5% level of significance is

Table 3: The computed the values of $\log L(\hat{\underline{\Omega}}; x)$, $\log L(\hat{\underline{\Omega}}^*; x)$ and the generalized likelihood ratio test statistic under $H_0 : \theta_2 = 0$.

	$\log L(\hat{\underline{\Omega}}^*; x)$	$\log L(\hat{\underline{\Omega}}; x)$	Test statistic
Data set 1	-201.807	-193.519	16.576
Data set 2	-386.387	-375.443	21.888

3.84 at one degree of freedom, the null hypothesis is rejected in both cases.

6. Simulation

It is difficult to compare the theoretical performance of estimators of different parameters of the GALSD obtained by the method of maximum likelihood. So in this section we have carried out a simulation study for comparing the performance of estimators obtained by the method of maximum likelihood. We have simulated two data sets for the following two sets of parameters: (i) $\theta_1 = 0.1501, \theta_2 = 0.1131, \alpha = 0.1928$ (underdispersion) and (ii) $\theta_1 = 0.4512, \theta_2 = 0.0981, \alpha = 0.3916$ (overdispersion). The computed values of the bias and standard errors of each of the estimators are presented in Table 4. From Table 4 it can be observed that both the bias and standard errors of the

Table 4: Bias and standard errors (with in parenthesis) of the estimators of the parameters θ_1, θ_2 and α of the GALSD for the simulated data sets.

Data set	Sample size	Method of maximum likelihood		
		$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\alpha}$
1	150	0.0480	0.0390	0.0299
		(0.1476)	(0.1470)	(0.1204)
	250	0.0212	0.0229	0.0163
		(0.1237)	(0.1400)	(0.1082)
	500	0.0116	0.0133	0.0127
		(0.1077)	(0.1005)	(0.1010)
	1000	0.0102	0.0118	0.0105
		(0.0951)	(0.0752)	(0.0701)
	1500	0.0050	0.0032	0.0050
		(0.0883)	(0.0714)	(0.0648)
2	150	0.0596	0.0405	0.0499
		(0.1456)	(0.2010)	(0.1548)
	250	0.0406	0.0300	0.0297
		(0.1249)	(0.1158)	(0.1118)
	500	0.0294	0.0283	0.0201
		(0.1063)	(0.1044)	(0.1005)
	1000	0.0197	0.0205	0.0155
		(0.0831)	(0.0761)	(0.0624)
	1500	0.0097	0.0120	0.005
		(0.0768)	(0.0632)	(0.0600)

estimators of the parameters θ_1, θ_2 and α of the GALSD show decreasing trend in their magnitude while the sample size increases from $n = 150$ to $n = 1500$.

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